X(3872) Analysis Update

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Fewer trees reduce the AUC gap, at a performance cost

Trees	Train AUC	Test AUC	ΔAUC	$\epsilon_B^{\text{train}}$	ϵ_B^{test}	ϵ_B^{test}
300	0.88451	0.85320	0.03131	2.890	4.082	7.055
500	0.89887	0.85645	0.04242	2.523	3.915	6.584
750	0.91212	0.85842	0.05370	2.114	3.809	6.872
1000	0.92145	0.86018	0.06126	1.803	3.558	6.690

Background efficiencies in percent: the first two columns use $\epsilon_X = 40\%$; the last uses 50%.

- 300 to 1000 trees: both train and test AUC increase.
- At 40% signal efficiency, later trees still reduce the test background.
- At 50%, the test background is not monotonic: 500 trees outperform 1000 here.

Reducing the gap alone does not identify the best ML configuration.

Same candidate split and seed; no retraining or independent split validation.

Hyperparameter variations do not resolve the AUC discrepancy

v1: best ML configuration in the 260916 update; hyperparameter tests to address overtraining.

Model	Change from v1	Train AUC	Test AUC	ΔAUC
v1	Baseline	0.92145	0.86018	0.06126
v2	$\lambda = 12$	0.92054	0.85907	0.06148
v3	$\alpha = 2$	0.92061	0.85916	0.06145
v4	min_child_weight $= 16$	0.91997	0.85878	0.06119
v5	$\gamma = 2$	0.92171	0.86024	0.06147
v6	max_depth $= 3$	0.89721	0.85482	0.04239

All six models: 1000 trees, identical samples and split. v1 depth $= 4$.

- Stronger regularization (v2–v5): little change in the gap.
- Shallower trees (v6): smaller gap, but lower test AUC.

Interim conclusion: hyperparameters are not the main explanation; inspect the MC weight components.

The $\hat{p}_T = 5$ component dominates weight, not statistics

$\hat{p}_T$	Weight	$N_{\text{train}}/N_{\text{test}}$	$\sum w$ (all)	Weight share [%]	Train AUC	Test AUC
5	1.84751	611/216	1527.895	46.00/48.08	0.95415	0.83900
10	0.134021	6900/2232	1223.884	37.68/36.04	0.88141	0.86390
15	0.0192029	19536/6424	498.507	15.29/14.86	0.91778	0.91163
30	0.000633152	37384/12545	31.613	0.96/0.96	0.97930	0.97659
50	5.9176e-05	28881/9508	2.272	0.07/0.07	0.98593	0.98281

Weight share shown as train/test; $\sum w = (N_{\text{train}} + N_{\text{test}})w$. AUC uses 1000 trees.

The five component labels follow the supplied $\hat{p}_T$ mapping; weights descend from 5 to 50.

Only 611 train + 216 test candidates in $\hat{p}_T = 5$, yet about 46–48% of MC weight.

A weighted sum is not a count of independent simulated events.

AUC by MC component and number of trees

Each cell: AUC(train) / AUC(test)

$\hat{p}_T$	300 trees	500 trees	750 trees	1000 trees
5	0.891 / 0.830	0.917 / 0.834	0.939 / 0.837	0.954 / 0.839
10	0.863 / 0.858	0.869 / 0.861	0.876 / 0.863	0.881 / 0.864
15	0.910 / 0.907	0.913 / 0.909	0.916 / 0.911	0.918 / 0.912
30	0.978 / 0.977	0.979 / 0.977	0.979 / 0.977	0.979 / 0.977
50	0.985 / 0.983	0.986 / 0.983	0.986 / 0.983	0.986 / 0.983

Same trained model, truncated at each tree count; no retraining.

The growing train/test gap is concentrated in the $\hat{p}_T = 5$ component.

Removing $\hat{p}_T = 5$: AUC comparison and conclusions

- **Original configuration:** AUC(train/test) = **0.921 / 0.860**.
  Train and test both include all five MC components.
- **Retrained without $\hat{p}_T = 5$:** AUC(train/test) = **0.904 / 0.859**.
  Train: actual training sample, excluding $\hat{p}_T = 5$;
  test: unchanged original sample, including $\hat{p}_T = 5$.
- **Overtraining:** low raw statistics and high weights in $\hat{p}_T = 5$ drive the train/test AUC gap.
- **Variable-selection sample:** for $\texttt{Bpt} > 15$ GeV, treat the $\hat{p}_T = 5$ contribution as approximately negligible; omit it from training, testing and the MC efficiency reference.

Allowed inputs and the common selection

Group	Allowed inputs
Muon	`Bmu1pt`, `Bmu2pt`, `Bmu1eta`, `Bmu2eta`
Track	`Btrk1Pt`, `Btrk2Pt`, `Btrk1Eta`, `Btrk2Eta`
Track geometry	`Btrk1dR`, `Btrk2dR`, `BtrkPtimb`
Composite kinematics	`Bujeta`, `Btktkpt`, `Btktketa`
Vertex / pointing	`Bchi2Prob`, `BujvProb`, `BtktkvProb`, `Balpha`, `Bcos_dtheta`

Cut before ML: $15 < \texttt{Bpt} < 50$ GeV, $|\texttt{By}| < 1.6$, $\texttt{BQvalue} < 0.15$ GeV; no track ΔR cut.
Background: $3.80 < m < 3.85$ or $3.89 < m < 3.94$ GeV.
Cut after ML: additionally $0 < \texttt{Btrk1dR}, \texttt{Btrk2dR} < 0.35$.

AUC ranks candidates; mass-spectrum inspection selects the recipe

1. Start with `Btrk1dR` and `Btrk2Pt`; keep common samples, split and hyperparameters fixed.
2. Add each allowed variable and rank by **test AUC** and **AUC**.
3. Inspect DATA mass spectra at the same weighted MC efficiency: background mass dependence and changes under tighter cuts.
4. Select the next recipe using both inputs; do not automatically take the largest AUC.

AUC: combine train and test candidates, scan the score threshold to build the weighted ROC curve, and integrate its area. Signal MC uses `pThatreweight`; background has unit weight. **test AUC:** the same calculation using test candidates only.

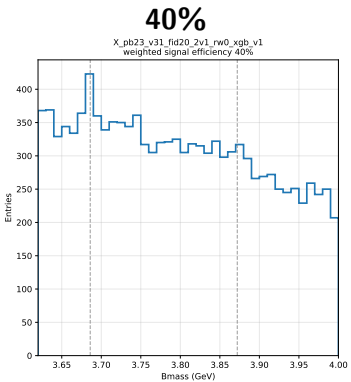
Recipe labels: *NvK* denotes *N* inputs and a recipe variant *K*.

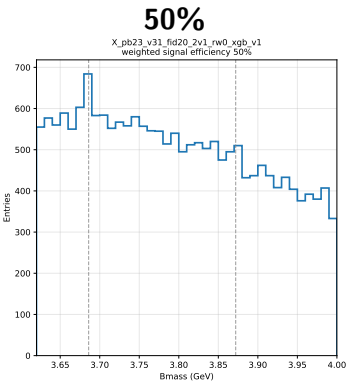
$$\textbf{2v1} \rightarrow \textbf{3v5} \rightarrow \textbf{4v35} \rightarrow \textbf{6v13} \rightarrow \textbf{7v17}$$

Mass-shape and efficiency concerns below are recorded human judgments, not completed bias or systematic studies.

2v1: DATA mass spectra at 40% and 50% MC efficiency

Inputs: `Btrk1dR`, `Btrk2Pt`





AUC: 0.84197; test AUC: 0.83291; train AUC: 0.84498.

Original selected DATA spectra; efficiencies refer to weighted MC after the application selection.

2v1: next-input ranking and the selection decision

Top eight additions by test AUC + baseline. Ranks use unrounded AUC in the full round.

Recipe	Added input	AUC	Rank	test AUC	Rank
3v3	`Btrk1Pt`	0.90450	1	0.89177	1
3v13	`Btktkpt`	0.90077	2	0.88902	2
3v4	`Btrk2dR`	0.89459	3	0.88250	3
3v5	`BtrkPtimb`	0.88780	4	0.87474	4
3v18	`Balpha`	0.85722	5	0.84166	5
3v15	`Bchi2Prob`	0.85577	6	0.84055	6
3v19	`Bcos_dtheta`	0.85516	7	0.83901	7
3v1	`Bmu1pt`	0.85379	9	0.83807	8
2v1	**Baseline (self)**	0.84197	18	0.83291	15

Choose 3v5: add `BtrkPtimb` (test AUC / AUC rank 4 / 4).

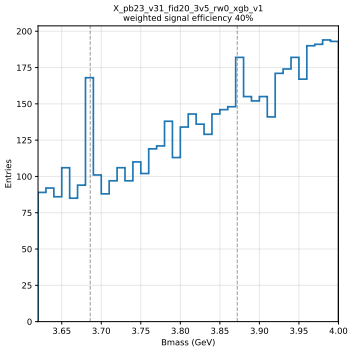
Higher ranks: `Btrk1Pt`, `Btktkpt` show positive background–mass association; `Btrk2dR` negative association.

Initial trigger: the background mass trend in 3v3; later judgments support the other exclusions.

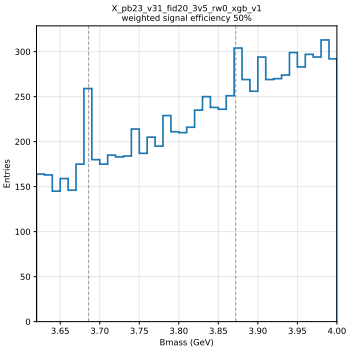
3v5: DATA mass spectra at 40% and 50% MC efficiency

Inputs: `Btrk1dR`, `Btrk2Pt`, `BtrkPtimb`

40%



50%



AUC: 0.88780; test AUC: 0.87474; train AUC: 0.89214.

Original selected DATA spectra; efficiencies refer to weighted MC after the application selection.

3v5: next-input ranking and the selection decision

Top eight additions by test AUC + baseline. Ranks use unrounded AUC in the full round.

Recipe	Added input	AUC	Rank	test AUC	Rank
4v22	`Btrk1Pt`	0.90593	1	0.89173	1
4v29	`Btktkpt`	0.90419	2	0.89070	2
4v23	`Btrk2dR`	0.90205	3	0.88822	3
4v34	`Balpha`	0.89775	4	0.88307	4
4v31	`Bchi2Prob`	0.89591	6	0.88171	5
4v35	`Bcos_dtheta`	0.89609	5	0.88155	6
4v20	`Bmu1pt`	0.89269	8	0.87786	7
4v21	`Bmu2pt`	0.89298	7	0.87781	8
3v5	**Baseline (self)**	0.88780	17	0.87474	17

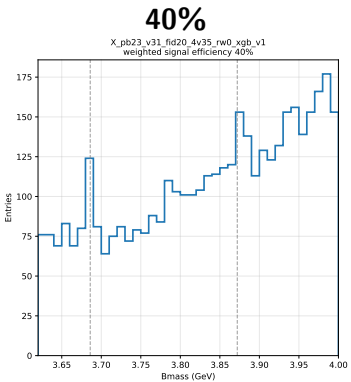
Choose 4v35: add `Bcos_dtheta` (test AUC / AUC rank 6 / 5).

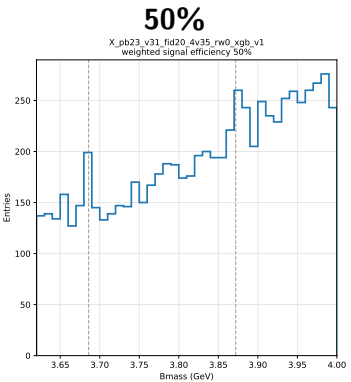
Ranks 1–3: background mass dependence. `Balpha` and `Bchi2Prob`: unstable peak changes under tighter cuts.

Selection favors background shape and cut stability; no fit-bias claim.

4v35: DATA mass spectra at 40% and 50% MC efficiency

Inputs: `Btrk1dR`, `Btrk2Pt`, `BtrkPtimb`, `Bcos_dtheta`





AUC: 0.89609; test AUC: 0.88155; train AUC: 0.90089.

Original selected DATA spectra; efficiencies refer to weighted MC after the application selection.

4v35: next-input ranking and the selection decision

Top eight additions by test AUC + baseline. Ranks use unrounded AUC in the full round.

Recipe	Added input	AUC	Rank	test AUC	Rank
5v9	`Btrk1Pt`	0.91157	1	0.89646	1
5v16	`Btktkpt`	0.90960	2	0.89481	2
5v10	`Btrk2dR`	0.90743	3	0.89234	3
5v18	`Bchi2Prob`	0.90219	4	0.88678	4
5v21	`Balpha`	0.89953	5	0.88383	5
5v7	`Bmu1pt`	0.89928	7	0.88335	6
5v8	`Bmu2pt`	0.89935	6	0.88299	7
5v20	`BtktkvProb`	0.89805	14	0.88205	8
4v35	**Baseline (self)**	0.89609	16	0.88155	13

Choose 6v13: add `Bmu1pt` and `Bmu2pt` together.

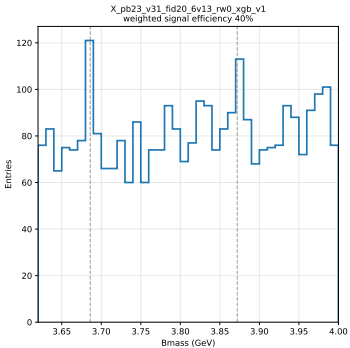
Ranks 1–3: background mass dependence; ranks 4–5: strict-cut anomalies, suspected MC efficiency limitations.

Two-input step approved separately; 6v13 is outside this one-input ranking.

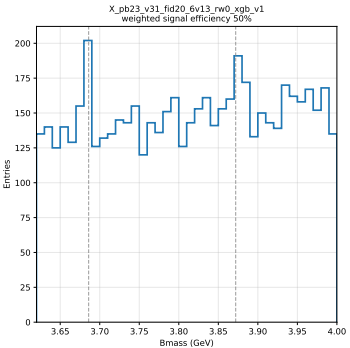
6v13: DATA mass spectra at 40% and 50% MC efficiency

Inputs: `Btrk1dR`, `Btrk2Pt`, `BtrkPtimb`, `Bcos_dtheta`
`Bmu1pt`, `Bmu2pt`

40%



50%



AUC: 0.90434; test AUC: 0.88736; train AUC: 0.90995.

Original selected DATA spectra; efficiencies refer to weighted MC after the application selection.

6v13: next-input ranking and the selection decision

Top eight additions by test AUC + baseline. Ranks use unrounded AUC in the full round.

Recipe	Added input	AUC	Rank	test AUC	Rank
7v11	`Btrk1Pt`	0.91799	1	0.90093	1
7v18	`Btktkpt`	0.91637	2	0.89907	2
7v12	`Btrk2dR`	0.91451	3	0.89812	3
7v20	`Bchi2Prob`	0.90929	4	0.89192	4
7v23	`Balpha`	0.90711	5	0.88946	5
7v17	`Bujeta`	0.90601	8	0.88793	6
7v15	`Btrk1Eta`	0.90564	11	0.88779	7
7v13	`Bmu1eta`	0.90637	6	0.88778	8
6v13	**Baseline (self)**	0.90434	14	0.88736	11

Choose 7v17: add `Bujeta` (test AUC / AUC rank 6 / 8).

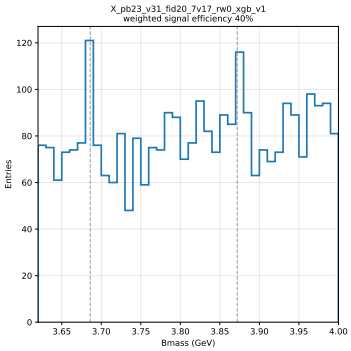
Same exclusions: `Btrk1Pt`/`Btktkpt` positive, `Btrk2dR` negative background–mass association; `Bchi2Prob`/`Balpha` strict-cut anomalies.

The mass-spectrum judgment takes priority over AUC rank.

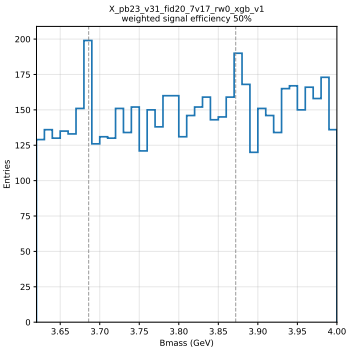
7v17: DATA mass spectra at 40% and 50% MC efficiency

Inputs: `Btrk1dR`, `Btrk2Pt`, `BtrkPtimb`, `Bcos_dtheta`
`Bmu1pt`, `Bmu2pt`, `Bujeta`

40%



50%



AUC: 0.90601; test AUC: 0.88793; train AUC: 0.91198.

Original selected DATA spectra; efficiencies refer to weighted MC after the application selection.

7v17: next-input ranking and the selection decision

Top eight additions by test AUC + baseline. Ranks use unrounded AUC in the full round.

Recipe	Added input	AUC	Rank	test AUC	Rank
8v23	`Btrk1Pt`	0.91913	1	0.90079	1
8v29	`Btktkpt`	0.91765	2	0.89927	2
8v24	`Btrk2dR`	0.91552	3	0.89829	3
8v31	`Bchi2Prob`	0.91070	4	0.89231	4
8v34	`Balpha`	0.90903	5	0.89031	5
8v30	`Btktketa`	0.90672	8	0.88843	6
8v25	`Bmu1eta`	0.90710	7	0.88818	7
8v32	`BujvProb`	0.90724	6	0.88793	9
7v17	**Baseline (self)**	0.90601	13	0.88793	8

Retain 7v17: no additional input is adopted.

Ranks 1–5 retain the preceding concerns. `Btktketa` additionally suppresses the observed X feature, raising an MC-efficiency concern.

Remaining test AUC gains do not motivate another step in the human choice.

Interim conclusion: 7v17 is the preferred recipe in this study

Recipe	train AUC	test AUC	AUC
2v1	0.84498	0.83291	0.84197
3v5	0.89214	0.87474	0.88780
4v35	0.90089	0.88155	0.89609
6v13	0.90995	0.88736	0.90434
7v17	0.91198	0.88793	0.90601

- AUC gains are considered together with background mass shape and stability under tighter cuts.
- The selected seven inputs are `Btrk1dR`, `Btrk2Pt`, `BtrkPtimb`, `Bcos_dtheta`, `Bmu1pt`, `Bmu2pt`, `Bujeta`.
- Further additions are not adopted; the current choice is **7v17**.

Preferred within the current allowed-variable study and MC assumption; final nominal selection and working point remain open.

Punzi optimization: resolve the relevant efficiency range

Henrique's suggestion: scan score cuts from 0.05 to 0.95 in steps of 0.05.

$$\mathrm{FOM}(t) = \frac{\epsilon_{\mathrm{MC}}(t)}{a/2 + \sqrt{B(t)}}, \qquad a = 5, \qquad B(t) = \frac{N_{\mathrm{left}}(t) + N_{\mathrm{right}}(t)}{3}.$$

- ϵ_{MC}: fraction of weighted MC passing score $> t$, within $3.62 < m < 4.00$ GeV; weight = `pThatreweight`.
- Signal window: $[3.84, 3.90]$ GeV; sidebands: $(3.75, 3.84)$ and $(3.90, 3.99)$ GeV. Window acceptance factor $= 1$.
- B: unit-weight DATA sideband counts, scaled by the window-width ratio; linear-background approximation.
- **xeff**: weighted reference-MC passing fraction without the mass cut; used to set the scan thresholds.

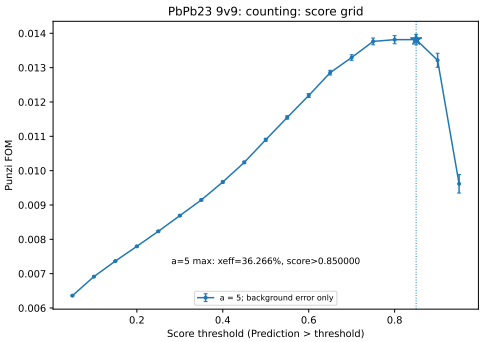
Fine scan: target xeff 15%–60% in 1% steps (46 points).

Coarse score scan: only 4 points in this efficiency range.

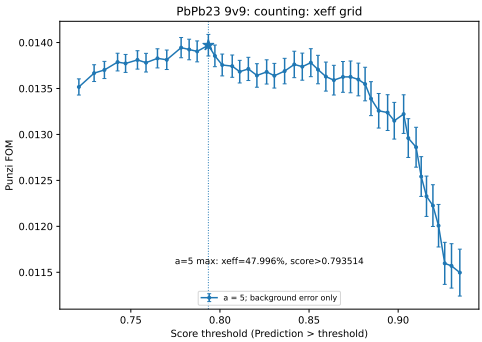
Shown errors include sideband counting statistics only. The 260916 $S/\sqrt{S+B}$ scan used a different definition.

9v9: coarse and fine Punzi scans versus score

Coarse score scan



Fine xeff scan

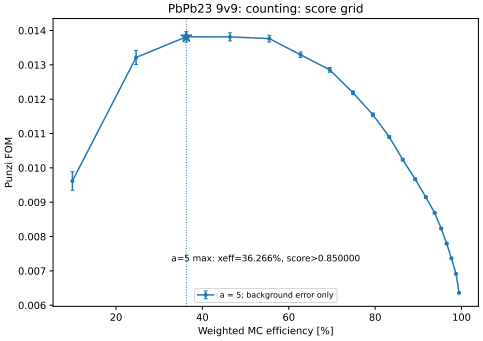


- Coarse-grid maximum: score 0.85, xeff 36.27%; agrees with Henrique's reported cut.
- Fine-grid maximum: score 0.793514, target xeff 48%; missed by the coarse grid.

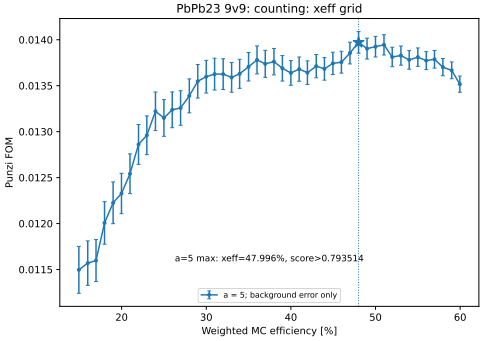
Agreement concerns the coarse-grid maximum; Henrique's full original settings are unavailable.

9v9: efficiency spacing explains the coarse-scan limitation

Coarse score scan



Fine xeff scan

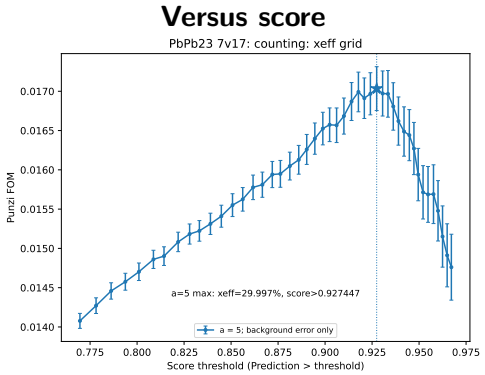


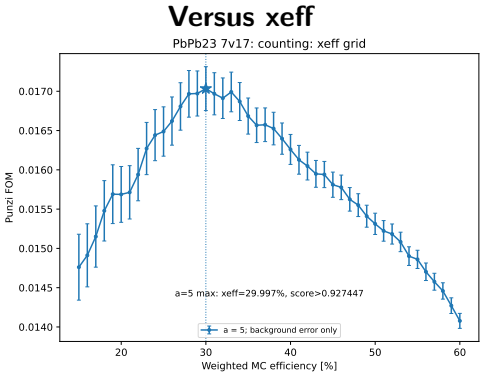
Coarse points in the chosen range: xeff 24.64, 36.27, 46.39, 55.49%.

Fine-scan maximum: 48%; points above 99% of peak FOM: 47–51%.

Interim conclusion: scan in xeff to resolve the relevant efficiency range and the Punzi plateau.

7v17: fine Punzi scan selects xeff 30%

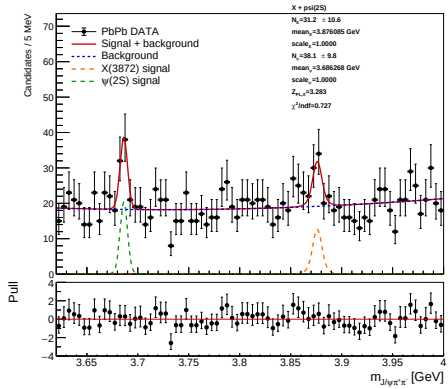




- **Optimal on this grid: xeff 30%, score** > 0.927447; FOM $= 0.017034$.
- Points above 99% of peak FOM: xeff 28%–34%; maximum is inside the grid.

Relative to 9v9: an additional track1 ΔR cut and removal of the pthat5 MC component. The shift is not attributable to variables alone.

7v17: fit at the Punzi-optimal cut, xeff 30%



score > 0.927447

$N_X = 31.17 \pm 10.59$
$N_{\psi(2S)} = 38.15 \pm 9.83$

MC widths fixed;
DATA means float independently;
C2 background.

Profile $\sqrt{q_0} = 3.283$

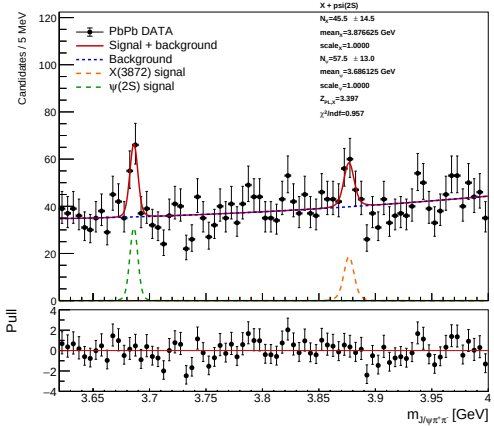
Interim conclusion: use the 30% grid optimum for the fit display; final nominal working point remains open.

Thanks for listening!

Reporter: Lekai Yao

2026.9.20

Backup: 7v17 fit at xeff 40%



score > 0.890241

$N_X = 45.50 \pm 14.53$
$N_{\psi(2S)} = 57.51 \pm 13.02$

Same fit model as at 30%.

Profile $\sqrt{q_0} = 3.397$

40% is an existing reference fit, not the systematic Punzi optimum. Shown only as backup; the grid optimum remains 30%.